Report

Assignment: lane detection using opencv

1. developed a lane line extractor using OpenCV and Python. The assignment processes static road images to identify highway lane markings using computer vision techniques.

2. Work Completed

- Loaded the image
- Implemented HSV color filtering to identify yellow and white lane markings.
- Applied Gaussian blur method to reduce image noise.
- Used Canny edge detection to identify edges in the road lane in the image.
- Applied Region of Interest (ROI) masking to focus on the road area in the given image.
- Used the Hough Transform to detect line segments from the img and draw them on the original image.

3. Tools and Technologies

- **Programming Language:** Python
- **Libraries:** OpenCV and NumPy
- **Methods I used:** HSV color filtering, Gaussian blur, Canny edge detection, ROI masking, and Hough Line Transform
- **Uploaded on** GitHub repository

4. Progress

The Python code and supporting assignment documentation have been prepared. The next step is to run the script with a suitable road image named road3.jpg present in my folder which has been taken from google and verify the extracted edges and detected lane lines from the uploaded image.

5. Challenges

- The script requires the input image to be present in the correct folder with the expected filename.
- Lane detection results may vary, this depends on image quality, lighting, and road conditions.

- Actual output screenshots still need to be generated and added to the GitHub repository.

6. I have uploaded the python code and README to the github repository

7. the output I got is making progress and it is able to detect the side line of the road and able to mark them using a colour line and able to find the road area in photo with curved path but in straight lines manner

Overall Status: In Progress